OPENSOURCESDRLAB DOCUMENTATION

# RFSoC-27DR Quick Start Guide

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|---|---|
| Product | RFSoC-27DR |
| Document | OSSDRLAB-RFSoC-27DR-QSG |
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## 1. Overview

The RFSoC-27DR is an 8-transmit / 8-receive development board built around the AMD Zynq UltraScale+ RFSoC XCZU27DR. Its eight RF-ADC inputs and eight RF-DAC outputs are on MMCX connectors; a TI LMK04828 provides the converter reference clocks and SYSREF. The board also has PS and PL DDR4, a QSFP28 cage, USB 3.0, Gigabit Ethernet, a microSD slot and a 120-pin expansion connector.

This guide shows how to run the three reference designs in this repository:

- **AD/DA loopback**: the RF-DACs play a test tone that is received by the RF-ADCs and shown in the Vivado logic analyser (ILA). The same Vitis workspace also contains a PS DDR4 diagnostics test, an Ethernet echo server and a USB 3.0 mass-storage example.
- **GTY IBERT**: a 25 Gb/s link test on the QSFP28 cage.
- **PL DDR4 test**: a write/read test of the PL DDR4.

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## 2. What you need

| Item | Notes |
|---|---|
| RFSoC-27DR board | with the fan connected |
| 12 V DC power supply | 5.5 × 2.1 mm barrel plug |
| USB Type-C cable | to the JTAG/UART port (FT2232H) |
| MMCX cables | to connect an RF-DAC output to an RF-ADC input |
| microSD card | FAT32, for booting from SD |
| Ethernet cable, USB 3.0 Type-C cable, QSFP28 loopback module or cable | only for the Ethernet, USB and GTY tests |
| PC with AMD Vivado and Vitis 2023.2 | Hardware Manager for JTAG and ILA; Vitis to run or rebuild the applications. Download from AMD: https://www.amd.com/en/support/downloads/adaptive-socs-and-fpgas/development-tools/2023-2.html |
| Serial terminal program | any terminal, 115200 baud, 8N1 |
| Prebuilt files | the three `*_prebuilt_Vivado2023.2.zip` files from the Releases page |
| AMD RFSoC Frequency Planner (optional) | to plan sample rates, NCO frequencies and Nyquist zones. Download from AMD: https://www.amd.com/en/products/adaptive-socs-and-fpgas/soc/zynq-ultrascale-plus-rfsoc.html |

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## 3. Board connectors and switches

| Connector / control | Function |
|---|---|
| RF-ADC inputs (8 × MMCX) | ADC tiles 224–227, two channels each |
| RF-DAC outputs (8 × MMCX) | DAC tiles 228–229, four channels each |
| CLK_IN, SYSREF_IN, TRIG_IN (MMCX) | external reference clock, external SYSREF, trigger input |
| SYS_CLKOUT, TRIG_OUT (MMCX) | clock output, trigger output |
| QSFP28 | GTY quad 128, up to 4 × 25 Gb/s |
| USB 3.0 Type-C | PS USB 3.0 (PS-GTR lane 0) |
| RJ45 | PS Gigabit Ethernet (RTL8211F) |
| USB Type-C (JTAG/UART) | JTAG and the PS UART0 console through an FT2232H |
| microSD slot | PS SD1, boot device |
| FX10A-120S-SV expansion connector | PL LVDS I/O (banks 88/89), PS MIO, PS-GTR lanes 1–3, GTY quad 129 |
| Boot-mode switch (PS_MODE[3:0]) | JTAG `0000`, QSPI `0010`, eMMC `0110`, SD `1110` |
| 12 V DC jack | board power |
| LED0–LED3 | user LEDs driven by the PS through EMIO |
| Fan header | 12 V fan for the RFSoC heat sink |

See `hardware/RFSoC-27DR_Schematic.pdf` for the schematic.

## 4. AD/DA loopback test

The design runs all eight RF-ADCs and eight RF-DACs at **4.096 GSPS** from a 256 MHz reference generated by the LMK04828. The application sets every RF-DAC and RF-ADC fine mixer (NCO) to **1000 MHz** and lets the PL generate a **10 MHz** tone (DDS), so each RF-DAC output carries a tone 10 MHz away from 1000 MHz. After loopback, the RF-ADC mixes it back down and the ILA shows a 10 MHz signal on every channel.

### 4.1 Boot from the SD card

1. Unzip `RFSoC-27DR_AD-DA-Loopback_prebuilt_Vivado2023.2.zip`. In that folder build the boot image with `bootgen -arch zynqmp -image boot.bif -o BOOT.bin -w` (bootgen comes with

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Vivado/Vitis 2023.2; Vitis > Create Boot Image does the same), then copy `BOOT.bin` to the root of a FAT32 microSD card.

2. Insert the card, set the boot-mode switch to **SD (`1110`)** and connect an RF-DAC output to an RF-ADC input with an MMCX cable.

3. Connect the JTAG/UART Type-C port to the PC, open the second (UART) serial port of the FT2232H at **115200 baud**, then power the board from 12 V.

4. The console prints the start-up messages:

```
RFSOC-27DR-Test-Start
waiting lmk04828 locked......
read from lmk04828 pll1_lock=0x0, pll12_lock=0x1
...
ADC converter samplerate=4096000000Hz
...
RFSOC initial Successful
Xadc Setup Successful
RFSOC Temperature=...
```

Every `XRFdc_GetPLLLockStatus` line should read `2` (tile PLL locked; `1` means unlocked). The four user LEDs then blink once per second and the die temperature is printed.

### 4.2 Look at the received signal

1. Open Vivado 2023.2, start the **Hardware Manager** and connect to the board (Open Target > Auto Connect).

2. Select the FPGA and set the probes file to `rfsoc_top.ltx` from the prebuilt zip. Do **not** program the device again: the bitstream was already loaded from the SD card.

3. Open the `hw_ila_1` window and run the trigger. The probe holds, for each of the eight receive channels, the first I and Q sample of every clock (`adc_tdata_1i` … `adc_tdata_8q`), followed by the eight DDS samples sent to the DACs.

4. Set the display of the ADC probes to **Analog** (right-click > Waveform Style > Analog) to see the 10 MHz sine wave on the looped-back channel.

> **Tip:** to run over JTAG instead of the SD card, set the boot-mode switch to JTAG (`0000`), program `rfsoc_top.bit` in Vitis or the Hardware Manager, then download and run `AD_DA_LOOP_MTS.elf` from Vitis (Run Configurations > Single Application Debug, using the `rfsoc_top.xsa` platform).

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### 4.3 Other PS tests in the same workspace

These applications use the same hardware platform ( `rfsoc_top.xsa` ) and are run from Vitis over JTAG. Their `.elf` files are in the prebuilt zip.

| Application | What it does |
|---|---|
| `DDR4_Test.elf` | AMD Zynq UltraScale+ MPSoC DRAM diagnostics. A menu on the serial console runs memory and eye-scan tests of the 2 GB PS DDR4. It runs from on-chip memory; see `ZYNQMP_DRAM_DIAGNOSTICS_TEST.docx` in `DDR4_Test/src/` . |
| `LWIP_Echo_Test.elf` | lwIP TCP echo server on the Gigabit Ethernet port. The console prints the board IP address (static default `192.168.1.10` ); data sent to TCP port 7 is echoed back. |
| `xusb_poll_example_1.elf` | USB 3.0 device example: the board appears on the PC as a small RAM-based mass-storage drive on the USB 3.0 Type-C port. |

## 5. GTY IBERT test (QSFP28)

1. Plug a QSFP28 loopback module, or a QSFP28 cable to other equipment, into the cage.
2. Unzip `RFSoC-27DR_GTY-IBERT-25G_prebuilt_Vivado2023.2.zip` , open the Vivado Hardware Manager with the boot-mode switch at JTAG, and program `example_ibert_ultrascale_gty_0.bit` with the probes file `example_ibert_ultrascale_gty_0.ltx` .
3. Create links for the four GTY channels of **quad 128** (Create Links in the Serial I/O Links tab). The design runs at **25 Gb/s** with a 156.25 MHz reference clock (MGTREFCLK0 of quad 128).
4. Reset the links and check that they come up and the error count stays at zero; run an eye scan if needed.

## 6. PL DDR4 test

1. Unzip `RFSoC-27DR_PL-DDR4-Test_prebuilt_Vivado2023.2.zip` and program `top.bit` with the probes file `top.ltx` in the Hardware Manager (boot-mode switch at JTAG).
2. In the ILA window, check that `init_calib_complete` is `1` (memory calibration passed) and trigger on `error` : it must stay `0` while the test writes the address as data across the 32-bit, 1 GB PL DDR4 (DDR4-2400) and reads it back.

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## 7. Reference designs in this repository

| Folder | Contents |
|---|---|
| `vivado/AD_DA_Loopback/AD_DA_LOOP_MTS_PL/` | Vivado 2023.2 project `AD_DA_LOOP_MTS_PL.xpr`: block design with the Zynq UltraScale+ PS, RF Data Converter and the `axi_lite_ctrl` register IP (in `repo/`); top level `rfsoc_top.v` with the DDS and ILA; constraints `rfsoc_27dr.xdc` |
| `vivado/AD_DA_Loopback/AD_DA_LOOP_MTS_PS/` | Vitis application sources (`*/src`) and the exported hardware platform `rfsoc_top.xsa` |
| `vivado/AD_DA_Loopback/lmk04828.tcs` | the LMK04828 configuration as a TI TICS Pro file; the application writes the same registers (`spi_ctrl.c`) |
| `vivado/GTY_IBERT_25G/ibert_ultrascale_gty_0_ex/` | IBERT UltraScale GTY example project |
| `vivado/PL_DDR4_Test/PL_DDR4_Test/` | PL DDR4 test project (DDR4 memory controller IP, test logic and ILA) |

Target device of all projects: `xczu27dr-fsve1156-2-i`.

To rebuild the AD/DA loopback design:

1. Open `AD_DA_LOOP_MTS_PL/AD_DA_LOOP_MTS_PL.xpr` in Vivado 2023.2. Vivado finds the `axi_lite_ctrl` IP through the project IP repository `repo/`. Generate the bitstream and export the hardware (File > Export > Export Hardware, include bitstream).
2. In Vitis 2023.2, create a platform project from the `.xsa` (operating system standalone, processor `psu_cortexa53_0`). For the Ethernet example add the `lwip211` library to the BSP (the prebuilt files use lwIP 2.1.1).
3. Create an empty application for each test and import the files of the matching `AD_DA_LOOP_MTS_PS/<application>/src` folder, including `lscript.ld`.
4. Build, then run over JTAG, or create `BOOT.bin` (Create Boot Image) with the FSBL, the bitstream and the application to boot from the SD card.

The GTY IBERT and PL DDR4 projects are opened in Vivado 2023.2 and rebuilt with Generate Bitstream.

## 8. Support

- Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-rfsoc-development-board-equipped-with-usb-30-socket
- Email: support@opensourcesdrlab.com

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